SLE-4: Final Viva & Architecture Decision Record (ADR)

Course: 02AML204 – Introduction to Artificial Intelligence | SY B.Tech. CSE (AI & ML)

PRN: 25UAM047

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System: AI Refrigerator Agent

PART A: Architecture Decision Record (ADR)

ADR Title: Use a Modular Architecture with a Separate Decision Engine

Status: Accepted **Date:** 12-10-2026

1. Context

The AI Refrigerator Agent monitors temperature, door status and item-condition information, then chooses actions such as cooling, reducing cooling, issuing an alert or making no change. A single block handling sensor input, state tracking, decisions, actions and alerts would be harder to understand and test. The SLE-3 C4 design separates these responsibilities into five containers: Input/Sensor Module, State Manager, Decision Engine, Action Controller and Alert/Output Module. The key decision is how to organize these responsibilities so the agent remains understandable and maintainable.

2. Decision

Use a modular architecture and keep the Decision Engine separate from sensor input, state management, action execution and user alerts. The Decision Engine evaluates the current refrigerator state and selects an action; other modules handle the data flow and execution.

3. Alternatives Considered

- **Alternative 1 – Monolithic design:** put input handling, rules and actions in one function or module.
- **Alternative 2 – Direct sensor-to-action flow:** let sensor readings trigger actions without maintaining a separate current-state model.
- **Alternative 3 – Modular design without a separate Decision Engine:** keep modules, but mix decision rules into the input or action-handling module.

4. Rationale / Justification

- The SLE-3 architecture assigns one clear responsibility to each of its five containers, making the system easier to explain and test.
- A separate State Manager lets decisions use the current refrigerator state rather than treating each sensor reading in isolation.
- Keeping the Decision Engine independent makes the rules easier to review or change without redesigning sensor collection or action execution.
- The Component-level design breaks decision-making into State Evaluator, Rule/Condition Checker, Goal/Safety Checker and Action Selector, giving each step a clear purpose.
- The SLE-2 profiling report compared BFS and DFS on a small directed graph: BFS averaged 0.003387 ms and expanded 9 nodes; DFS averaged 0.009296 ms and expanded 6 nodes across the reported experiment.
- Those SLE-2 figures are useful evidence that measured runtime and node expansions can tell different stories, but they are not direct benchmarks of the refrigerator architecture. They therefore should not be used to claim that this modular design is faster.
- Based on the SLE-3 design, modularity is chosen for clarity, separation of responsibilities and easier testing—not because an unmeasured performance improvement is assumed.

5. Consequences

Positive:

- Decision rules can be changed with less impact on input, state tracking and output modules.
- Each module can be explained and tested separately; the data flow is easier to follow.

Negative / Trade-offs:

- More modules create extra interfaces and require clear communication between them.
- The modular structure alone does not prove faster execution; real refrigerator data and further tests would be needed to measure performance.

6. References

- SLE-2: Profiling Report (BFS vs DFS comparison), PRN 25UAM047.
- SLE-3: Architectural Design (Full C4 Model) – AI Refrigerator Agent, PRN 25UAM047.

Note: The submitted SLE-2 report profiles BFS/DFS on a graph, while SLE-3 describes the AI Refrigerator Agent. This ADR keeps those pieces of evidence distinct and does not present the graph timings as refrigerator-agent performance results.

PART B: Short Summary of My SLE Journey

In SLE-1, my chosen project was an AI Refrigerator Agent. The agent receives information such as temperature, door status and item condition, evaluates the situation and chooses an appropriate action or alert. In SLE-2, I profiled Breadth-First Search (BFS) and Depth-First Search (DFS) on the same small directed graph, using 1,000 runs and recording runtime and nodes expanded. BFS measured 0.003387 ms and expanded 9 nodes; DFS measured 0.009296 ms and expanded 6 nodes. In SLE-3, I documented the refrigerator agent using the four C4 levels: Context, Container, Component and Code. The architecture separates input, state management, decision-making, action control and output. For SLE-4, I record the decision to keep the Decision Engine separate so its rules remain clear and independently testable.

PART C: AI Contribution Note

- **AI tools used:** ChatGPT.
- **What AI helped with:** Organizing the ADR, explaining alternatives and trade-offs, and aligning the document with the SLE-4 structure.
- **What I did myself:** Selected the AI Refrigerator Agent as the project, reviewed the proposed modules and design, and am responsible for understanding and defending the decision in the viva. I will verify that this description matches my actual implementation before submission.

PART D: Viva Preparation Notes

- **Why separate the Decision Engine?** It keeps the agent's decision rules independent from sensor input and action execution.
- **Why use a State Manager?** Decisions should consider the current refrigerator state, not only an isolated sensor reading.
- **What alternatives were rejected?** A monolithic design, direct sensor-to-action logic, and mixing decisions into other modules.
- **What did SLE-2 show?** BFS had the lower measured average runtime, while DFS expanded fewer nodes on that particular graph; the small test should not be over-interpreted.
- **What would be needed to claim better performance?** Test the refrigerator agent on representative inputs and compare results using the same conditions.